

UNIT 5

Computer Forensic Analysis & validation

Determining What Data to Collect and Analyse

1. Scope of Investigation

- Type of investigation (civil, criminal, corporate) decides what data needs to be collected.
- Civil cases → limited by court-ordered discovery.
- Criminal cases → limited by details in the search warrant.
- Corporate investigations → usually focus on policy violations (e.g., emails).

2. Collect Only Required Data

- Some cases require only a small set of specific items to speed up the process.
- But in corporate legal cases, investigators may be asked to collect as much data as possible.
- This increases effort, time, and cost.

3. Scope Creep

- Investigation may expand due to new unexpected evidence.

- Document every request for additional searches or extended work to justify time and cost later.

4. Criminal Investigations

- Defence can request complete discovery, sometimes revealing new evidence that helps the accused.
- Prosecutors must thoroughly examine all evidence before trial.

5. Data Collection Differences

- Corporate internal investigations → easier because data is readily accessible.
- Cyber-stalking or criminal cases → need cooperation from ISP or email services.

6. Data Retention Limits

- Some companies (e.g., AOL) have structures for investigation support.
- Many email services store emails only for 14–90 days.

7. Industrial Espionage Cases

- May require:
- Camera surveillance
- Software/hardware keyloggers
- Network monitoring
- Remote drive acquisition
- Checking if external devices were used

Evidence Collection Guidelines

8. Prepare Target Drives Properly

- Use only cleaned/wiped drives for imaging (e.g., Secure Wipe tools).

9. Document System Information

- List all physical components of the suspect's system.
- Note system state during seizure.

10. Check CMOS Date/Time

- Before static acquisition, verify system date/time.

11. Document Imaging Procedure

- Record how data was acquired.
- Generate hash values (MD5, SHA-1, etc.) for verification.

12. Analyse Data Systematically

- Review all files and folders.
- Tools like FTK can generate file/folder lists.

13. Start from the Root Directory

- Except in civil cases with limited search scope.

14. Recover Password-Protected Files

- Use tools like PRTK, Passware, NTI Password Recovery.

Validating Forensic Data

1. Importance of Validation

- Ensures integrity and authenticity of digital evidence.
- Required for presenting evidence in court.
- Prevents tampering, alteration, or corruption of data.

2. Use of Forensic Hashing

- Hashing verifies that the forensic image matches the original data.
- Common hashing algorithms:
 - MD5
 - SHA-1
 - (Better alternatives: SHA-256, SHA-512)
- Hash is generated when evidence is acquired and then compared during analysis.

3. Automatic Hashing in Forensic Tools

Most forensic tools perform hashing automatically:

- ProDiscover
- X-Ways Forensics
- FTK
- EnCase

Example:

ProDiscover recomputes the hash when an image file is loaded and compares it with the original acquisition hash.

4. Auto Image Verification

- Tools display an Auto Checksum Verification window when loading images.
- Confirms that no data has been changed since acquisition.

5. Limitations of Forensic Tools

- Some forensic suites cannot hash:
 - Specific sectors
 - Specific regions of the drive
 - Individual files selectively

Using Hexadecimal Editors for Validation

6. Role of Advanced Hex Editors

- Provide more control and functions than forensic tools.
- Can hash:
- Specific files
- Specific sectors or byte ranges
- Useful when searching for:
- Known illegal images
- Files with changed filenames but identical content.

7. Identical Content → Same Hash

- Even if filenames are changed:
- Same content = same hash
- Helps identify disguised or renamed files.

Addressing Data-Hiding Techniques & Remote Acquisitions

1. What is Data Hiding?

- Modifying files or systems to conceal information.
- Common techniques:
 - Hidden partitions
 - Hidden file properties
 - Changing file extensions
 - Bit-shifting
 - Encryption & passwords

Hiding Partitions

2. Methods of Hiding Partitions

- Create a partition → manually erase references using disk editors (e.g., Norton Disk Edit).
- Modify the partition table to remove visibility.

Partitions reappear when the table is restored and the system restarts.

- Use partitioning software to hide OS partitions:
 - GDisk
 - Partition Magic
 - System Commander
 - GRUB (Linux boot loader)
 - These hide other bootable partitions from the OS.

3. Importance During Forensic Analysis

- Always account for all disk space on evidence drives.
- Look for unallocated or unexplained space → could indicate:
 - Hidden partitions
 - Previously deleted partitions

4. Detecting Suspicious Partition Activity

- Check if partition sizes do not match:
 - Example:
 - Extended partition size = 5381.1 MB

- Visible logical drives (E + F) = 5271.3 MB
- Missing 109.8 MB → possible hidden partition

5. Indicators of Hidden Partitions

- Displayed as type “H” or “Hidden” in partition tables.
- Shown as:
 - Unknown partition in Windows Disk Manager
 - Highest drive letter (e.g., Z) in ProDiscover
- Non-sequential drive letters (e.g., skipping drive “I”) may indicate hidden drives.

6. Partition Gaps

- Windows typically creates small gaps between partitions:
 - Windows 2000/XP → ~63 sectors
 - Windows Vista → ~128 sectors
- Larger gaps (e.g., 109.8 MB) → suspicious; may contain hidden data.

7. Forensic Tools for Identifying Hidden Partitions

- ProDiscover → shows hidden partition with drive letter Z.
- FTK and others → can carve/salvage data from partition gaps.

Bit-Shifting

1. What is Bit-Shifting?

- A data-hiding technique where bits inside each byte of a file are rearranged.
- Makes data appear unreadable in text editors or word processors.
- Originally used by home computer users who wrote assembly/macro programs.

2. How Bit-Shifting Works

- A custom program scrambles the bits of a file containing sensitive or illegal data.
- A second program reverses the process to restore the original file.
- Still used today in simple encryption and data-hiding tools.

3. Impact on Forensic Investigations

- Bit-shifted files may appear as:
 - Random symbols
 - Corrupted data
 - Non-readable binary
- Can mislead investigators into thinking the file is useless or damaged.

4. Using Hex Workshop to Analyze Bit-Shifting

Steps:

- Open file (e.g., Bit shift.txt) in Hex Workshop.
- Enable Data Operations toolbar (Options → Toolbars).
- Use Shift Left operation to move bits left.
- Save output as Bit shift left.txt.
- Use Shift Right operation to restore original file.
- Save restored file as Bit shift right.txt.

5. Hash Validation After Bit-Shifting

- Generate MD5 hashes for:
 - Original file (Bit shift.txt)
 - Left-shifted file
 - Right-shifted (restored) file
- Use Tools → Generate Checksum → MD5 in Hex Workshop.
- Compare hash values:
 - Shifted files have different hashes
 - Restored file should match original hash

6. Relation to Malware Detection

- Antivirus tools also rely on hashing to detect malware.
- Some advanced malware uses bit-shifting to evade detection.
- Hex Workshop's manual bit-shifting can help uncover:
 - Hidden malware code
 - Bit-shifted payloads in Office files (Word, Excel)

7. Hidden Malware in Office Documents

- Some malicious Office documents store bit-shifted executable code at the file end.
- When opened, malware reverses the bit-shift → executes hidden code.

Performing Remote Acquisitions

1. What Are Remote Acquisitions?

- Imaging a computer's drive from a distant location.
- Useful when:
 - The computer is physically far away.
 - You don't want the suspect to know about the investigation.
- Saves time, cost, and travel efforts.

2. Advantages of Remote Acquisitions

- Covert – Suspect remains unaware.
- Cost-effective – No need to physically access the machine.
- Time-saving – Immediate imaging over the network.
- Ideal for large organizations or geographically distributed systems.

Remote Acquisitions with Runtime Software

3. Runtime Software Tools

Remote acquisition tools from Runtime Software (runtime.org):

- DiskExplorer for FAT
- DiskExplorer for NTFS
- HDHOST

4. DiskExplorer (FAT/NTFS)

- File system–specific tools.
- Can create:
 - Raw image files
 - Segmented image files
- Useful for disk imaging and archiving.

5. HDHOST – Remote Communication Tool

- Allows two computers to connect remotely.
- Used along with DiskExplorer (matching FAT or NTFS).
- Enables live, real-time remote acquisition of a target machine.

7. When to Use Remote Acquisition

- Covert investigations.
- Corporate environments with remote offices.
- Cybercrime cases where immediate imaging is required.
- Situations where physical access is restricted or risky.

Network Forensics Overview

1.Introduction to Network Forensics

- Process of collecting, analyzing, and tracking raw network traffic
- Helps determine how attacks occurred or how events unfolded
- Increasing importance due to rising network threats
- High demand for specialists — shortage of nearly 50,000 experts projected

2.Importance of Network Forensics

- Helps differentiate between actual attacks and system/application errors
- Prevents unnecessary time & money wastage in false-alarm situations
- Essential for post-attack investigation and evidence preservation

3.Role of Network Forensics Investigators

- Establish standard operating procedures (SOPs) for incident data collection
- Assist administrators in identifying compromised systems
- Analyze network logs, packets, and communication patterns

Keeping a Network Safe : -

4: Network Hardening Overview

- Essential to secure systems before an attack happens
- Mitigates risks from malware, viruses, intrusions, and vulnerabilities
- Uses network forensics to understand breach points and attack paths

5. Network Hardening Techniques

- Apply the latest security patches
- Implement layered defense strategy
- Hide critical data deeper in the network
- Increase security controls as network depth increases

Defense in Depth (DiD)

Defense in Depth (NSA Model)

- Developed by the National Security Agency (NSA)
- Multi-layered protection strategy
- Ensures if one layer fails, others still defend

Three Layers of DiD

1. People

- Hire skilled cybersecurity professionals
- Proper training on security policies
- Good workforce management to avoid insider threats

2. Operations

- Define clear security processes
- Physical security controls
- Incident handling protocols

3. Technology

- Strong network architecture
- Use of firewalls, IDS/IPS, VPNs
- Regular penetration testing & risk assessments

Technological Safeguards

- Deploy well-tested security solutions
- Maintain network monitoring & intrusion detection
- Rapid-response procedures for breach containment
- Log management for forensic visibility

Performing Live Acquisitions

1.What is Live Acquisition?

- Process of collecting data while the system is running
- Focus on capturing volatile data before it disappears
- Volatile data includes: RAM, active processes, network connections, system state

2.Order of Volatility (OOV)

- OOV = how long data persists on a system
- Highly volatile data (disappears first):
 - CPU registers & cache
 - RAM
 - Running processes
 - Network connections
- Less volatile data:
 - Hard drive data
 - Backups & archives (can last for years)

3. Steps for Live Acquisition

- Test your forensic bootable CD before using it on a suspect system
- If suspect system is network-connected → add required forensic tools to your workstation
- If remote access is not available → boot suspect machine with forensic CD
- Document every action and reason (chain of custody requirement)

4. Data Storage for Live Acquisition

- Best practice: save collected data to a network drive
- If network not available → connect a USB thumb drive
- Record the use of any external device in logs/journal

5. Copying Volatile Memory (RAM)

- RAM must be collected immediately due to high volatility
- Tools for RAM acquisition:
 - Microsoft built-in utilities
 - Back
 - Memfetch

6. Additional Actions Based on Case

- For intrusion investigations:
 - Use Rootkit Revealer to detect hidden malware
- After memory capture:
 - Create drive image over the network
 - OR shut down and perform static acquisition later
 - OR check firmware to detect tampering

7. Ensuring Integrity

- Generate forensically sound hash values for all collected files
- Guarantees evidence was not altered
- Common hash types: MD5, SHA-1, SHA-256

Performing Live Acquisition in Windows

8. Windows RAM Acquisition Tools

- ManTech Memory DD
 - Supports RAM acquisition up to 4 GB
- Win32dd
 - Freeware memory dumping tool
- Winen.exe (Guidance Software)
 - Suitable for beginners

Slide 9: BackTrack Linux for Live Acquisition

- Combines tools from:
 - White Hat Hackers CD
 - The Auditor CD
- Contains 300+ tools such as:
 - Forensic tools
 - Network sniffers
 - Password crackers
- Used widely in security competitions (like CCDC)
- Popular among penetration testers

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Using Network Tools

1.Introduction to Network Tools

- Network tools assist administrators & forensic investigators
- Used for:
 - Remote shutdowns
 - Monitoring device usage
 - Analyzing network traffic
 - Detecting harmful processes

Windows Operating System Network Tools

Tools	Description
RegMon	It Shows all registry data in real time
Process explorer	Shows what files, Registry keys, and dynamic link libraries (DLLs) are loaded at a specific time.
Handle	Shows what files are open and which processes are using these files.
Filemon	Shows file system activity.
Psexec	Runs processes remotely
Pstoolsid	Displays the security identifier (SID) of a computer or user
Pskill	Kills processes by name or process ID
Pstools	Lists detailed information about processes

Tools	Description
Pstools	Displays who's logged on locally
Pstools	Allows you to change account passwords
Pstools	Enables you to view and control services
Pstools	Shuts down and optionally restarts a computer
Pstools	Allows you to suspend processes

Linux Operating System Network Tools

Tools	Description
dcfldd	The U.S. DOD computer forensics lab version of the dd command
memfetch	Forces a memory dump
photorec	Retrieves files from a digital camera
snort	A popular IDS that performs packet capture and analysis in real time
oinkmaster	Helps manage snort rules so that you can specify what items to ignore as regular traffic and what items should raise alarms
john	The latest version of John the Ripper, a password cracker
chntpw	Enables you to reset passwords on a Windows computer, including the administrator password
tcpdump and ethereal	Packet sniffers

Performing Live Acquisitions

1.Introduction to the Honeynet Project

- Global research initiative focused on network security awareness.
- Aims to make information, tools, and research openly available.
- Helps organizations understand network threats and attacker behaviors.

Goals of the Honeynet Project

1.Awareness

- Educating individuals and organizations about existing cyber threats.

2. Protection

- Providing knowledge on:
 - How attackers operate
 - Communication methods
 - Techniques used for attacks

3. Research Support

- Offering tools, datasets, and methodologies for cybersecurity research.

Major Threats Addressed

Distributed Denial-of-Service (DDoS) Attacks

- Attackers use hundreds or thousands of compromised “zombie” machines.
- Major concerns:
 - Financial damage
 - Difficulty in tracing attack sources

Zero-Day Attacks

- Attackers exploit unknown vulnerabilities before patches are released.

Components of the Honeynet Project

Honeypots

- Fake computers designed to look like real network systems.
- Contain no valuable information.
- Purpose: lure attackers and observe their behavior without risk to actual systems.

Honeywalls

- Dedicated machines that monitor honeypots.
- Record attacker activity, tools, and techniques.

How Honeypots Work

- Not used for real network operations — act only as bait.
- Steps:
 - a. Load standard software used in the network.
 - b. Create a forensic image (baseline image).
 - c. Deploy honeypot on the network.
 - d. If compromised → take it offline.

- e. Create a new forensic image after attack.
- f. Compare both images to identify:
 - Attack methods
 - Files modified or added
- Both images stored in a database for research.

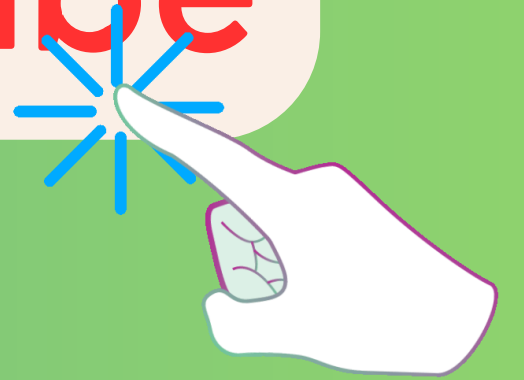
7: Benefits of the Honeynet Project

- Helps understand attacker intentions & techniques.
- Improves network defense strategies.
- Provides valuable real-world attack data.
- Reduces impact of emerging threats like DDoS & zero-day attacks.



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